Week-2

1. a) Create and attach an EBS volume to an EC2 instance, and
   b) scale the instance by increasing CPU and RAM by changing the instance type.

2.Attach and permanently mount an EBS volume to a Linux EC2 instance to ensure data persistence across reboots.

3. Create a snapshot of the attached EBS volume and use it to create and attach a new volume to an EC2 instance in another AWS region.

1. a) Create and attach an EBS volume to an EC2 instance, and
   b) scale the instance by increasing CPU and RAM by changing the instance type.

a)Creating andattaching an EBS volume to an EC2 instance

Step 1: Create a New EBS Volume

Open the EC2 Dashboard.

From the left-hand menu, select Volumes under Elastic Block Store (EBS).

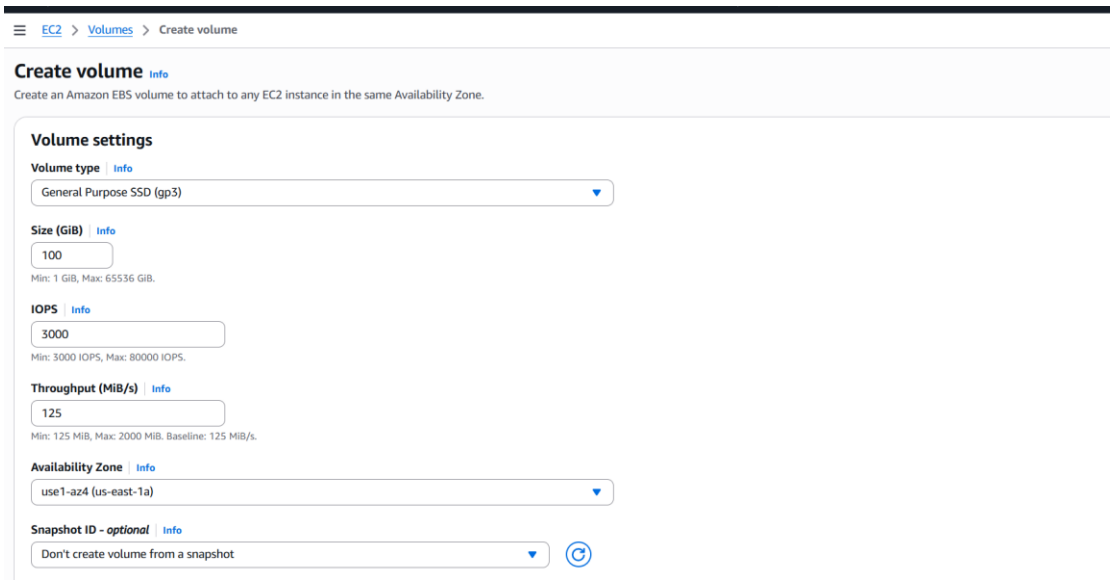
Click Create Volume.

Choose:

Volume type (gp3/gp2)

Size of the volume

Availability Zone (must be the same as the EC2 instance)



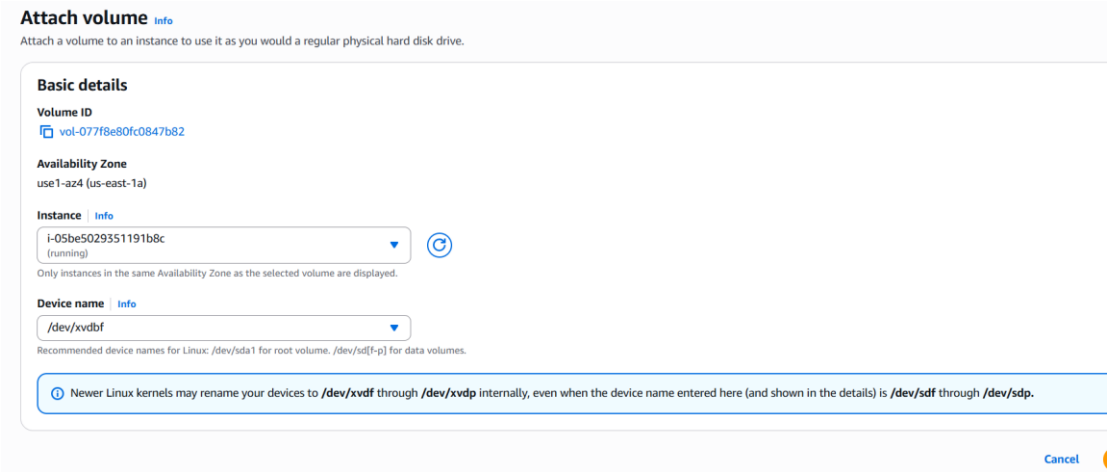
Step 2: Attach the EBS Volume to the Existing EC2 Instance

In the Volumes section, select the newly created volume.

Click Actions → Attach Volume.

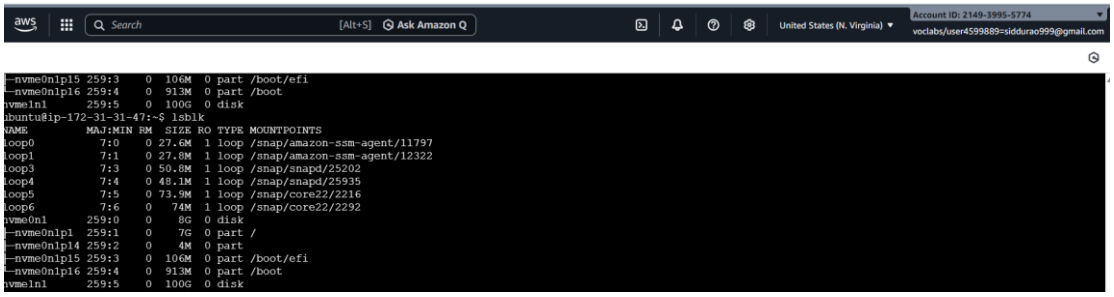
Select the target EC2 instance from the list.

Specify the device name (example: /dev/xvdf).
Click Attach.



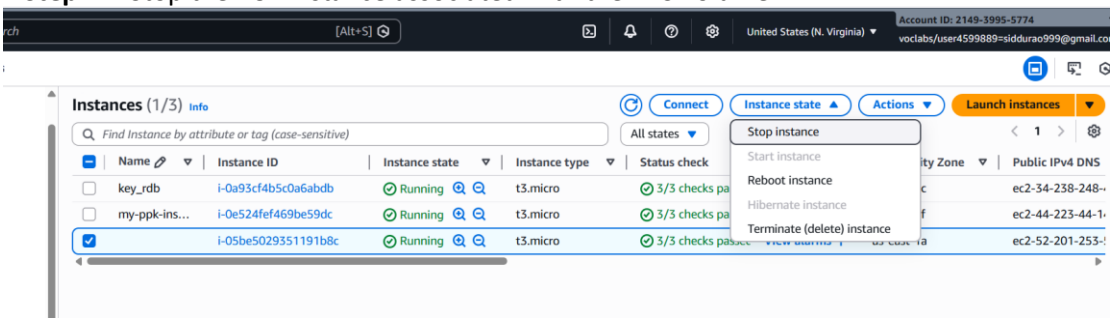
Step 3: Verify the Volume on the EC2 Instance
Connect to the EC2 instance using SSH.
Check the attached disks:
lsblk
Identify the new volume (e.g., /dev/xvdf).

Note: Ensure that the EBS volume and the EC2 instance belong to the **same Availability Zone**, as cross-AZ attachment is not supported.



b)Procedure: increasing/decreasing of CPU and RAM

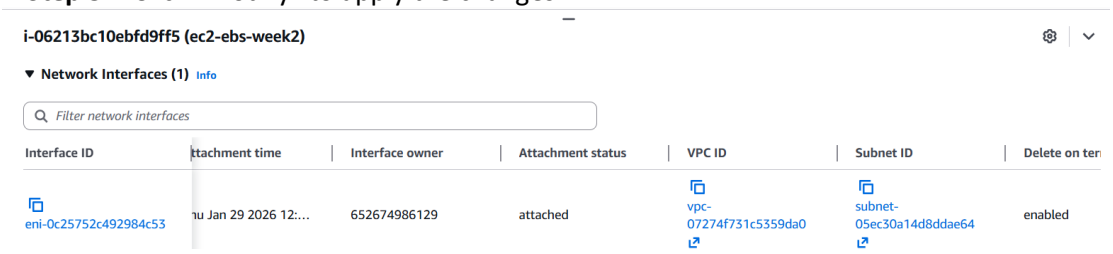
Step 1. - Navigate to the EC2 dashboard.
Step 2. - Stop the EC2 instance associated with the EBS volume.



Step 3. - Click "Actions" and then—instance settings—change the instance type"Modify Volume".

Step 4. -Increase/ decrease the size or change the volume type to a higher performance specification.

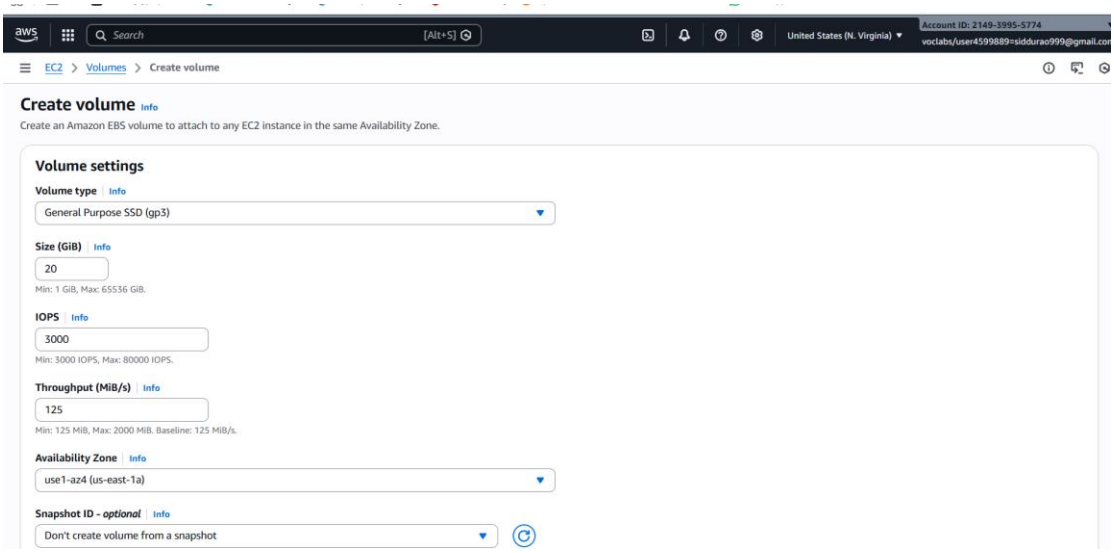
Step 5. - Click "Modify" to apply the changes.



2.Attach and permanently mount an EBS volume to a Linux EC2 instance to ensure data persistence across reboots.

Step 1: Create an EBS Volume

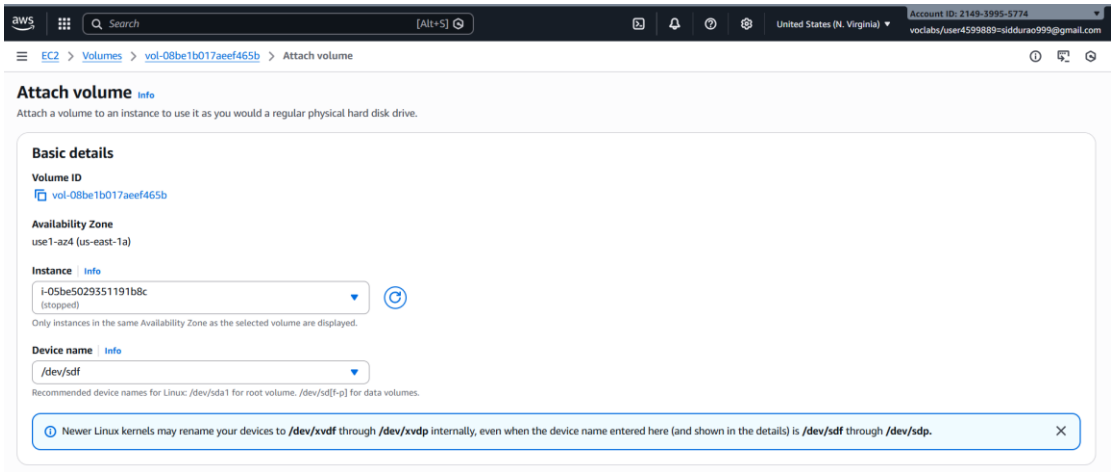
1. Log in to **AWS Management Console**
2. Go to **EC2 → Volumes**
3. Click **Create Volume**
4. Choose:
   - **Volume Type**: gp3 (recommended)
   - **Size**: (e.g., 20 GB)
   - **Availability Zone**: **Same AZ as EC2 instance**
5. Click **Create Volume**



Step 2: Attach the EBS Volume to EC2

1. Select the newly created **EBS volume**

2. Click **Actions → Attach Volume**
3. Choose:
    - **Instance**: Select your Linux EC2 instance
    - **Device name**: /dev/xvdf
4. Click **Attach**



Step 3: Connect to the EC2 Instance

Connect using SSH:

ssh -ikey.pem ec2-user@<public-ip>

Step 4: Verify the Attached Disk

List all block devices:

lsblk

You will typically see:

- Root volume: /dev/nvme0n1 (with partitions like nvme0n1p1)
- New EBS volume: /dev/nvme1n1 (**no partition yet**)

Check filesystem information:

lsblk –fs

```
loop0          7:0    0 27.6M  1 loop /snap/amazon-ssm-agent/11797
loop1          7:1    0 27.8M  1 loop /snap/amazon-ssm-agent/12322
loop2          7:2    0 73.9M  1 loop /snap/core22/2216
loop3          7:3    0 50.8M  1 loop /snap/snapd/25202
loop4          7:4    0 48.1M  1 loop /snap/snapd/25935
loop5          7:5    0   74M  1 loop /snap/core22/2292
nvme1n1      259:0    0   20G  0 disk
nvme2n1      259:1    0  100G  0 disk
nvme0n1      259:2    0    8G  0 disk
├─nvme0n1p1  259:3    0    7G  0 part /
├─nvme0n1p14 259:4    0    4M  0 part
├─nvme0n1p15 259:5    0  106M  0 part /boot/efi
└─nvme0n1p16 259:6    0  913M  0 part /boot
ubuntu@ip-172-31-31-47:~$ lsblk -fs
NAME        FSTYPE   FSVER LABEL          UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0       squashfs 4.0                                                           0   100% /snap/amazon-ssm-agent/11797
loop1       squashfs 4.0                                                           0   100% /snap/amazon-ssm-agent/12322
loop2       squashfs 4.0                                                           0   100% /snap/core22/2216
loop3       squashfs 4.0                                                           0   100% /snap/snapd/25202
loop4       squashfs 4.0                                                           0   100% /snap/snapd/25935
loop5       squashfs 4.0                                                           0   100% /snap/core22/2292
nvme1n1
nvme2n1
nvme0n1p1   ext4     1.0   cloudimg-rootfs 12f6092b-46d6-4b4c-8ac7-d76c4d62dde1    4.2G    37% /
└─nvme0n1
nvme0n1p14
└─nvme0n1
nvme0n1p15 vfat      FAT32 UEFI           BDC4-D6EC                              98.2M     6% /boot/efi
└─nvme0n1
nvme0n1p16 ext4      1.0   BOOT           3ad5394a-0d50-4795-835e-1b2da7ac508d  664.2M    18% /boot
└─nvme0n1
```

Step 5: Create a Partition

Create a partition on the new volume:

fdisk /dev/xvdf

Inside fdisk:

- Press m → Help menu
- Press n → New partition
- Press p → Primary partition
- Press Enter → Default partition number
- Press Enter → Default first sector
- Press Enter → Default last sector (use full disk)
- Press w → Write changes and exit

Notify kernel:

Partprobe

```
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x5c54de2a.

Command (m for help): m

Help:

  DOS (MBR)
   a   toggle a bootable flag
   b   edit nested BSD disklabel
   c   toggle the dos compatibility flag

  Generic
   d   delete a partition
   F   list free unpartitioned space
   l   list known partition types
   n   add a new partition
   p   print the partition table
   t   change a partition type
   v   verify the partition table
   i   print information about a partition

  Misc
```

Step 6: Verify the New Partition

lsblk -fs

You should now see:

/dev/xvdf1

```
ubuntu@ip-172-31-31-47:~$ lsblk -fs
NAME        FSTYPE   FSVER LABEL          UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0       squashfs 4.0                                                           0   100% /snap/amazon-ssm-agent/11797
loop1       squashfs 4.0                                                           0   100% /snap/amazon-ssm-agent/12322
loop2       squashfs 4.0                                                           0   100% /snap/core22/2216
loop3       squashfs 4.0                                                           0   100% /snap/snapd/25202
loop4       squashfs 4.0                                                           0   100% /snap/snapd/25935
loop5       squashfs 4.0                                                           0   100% /snap/core22/2292
nvme2n1
nvme0n1p1   ext4     1.0   cloudimg-rootfs 12f6092b-46d6-4b4c-8ac7-d76c4d62dde1    4.2G    37% /
└─nvme0n1
nvme0n1p14
└─nvme0n1
nvme0n1p15 vfat      FAT32 UEFI           BDC4-D6EC                              98.2M     6% /boot/efi
└─nvme0n1
nvme0n1p16 ext4      1.0   BOOT           3ad5394a-0d50-4795-835e-1b2da7ac508d  664.2M    18% /boot
└─nvme0n1
nvme1n1p1
└─nvme1n1
ubuntu@ip-172-31-31-47:~$
```

Step 7: Format the Partition

Format with XFS filesystem:

mkfs.xfs /dev/xvdf1

Verify:

lsblk -fs

```
ubuntu@ip-172-31-31-47:~$ sudo mkfs.xfs -f /dev/nvme1n1
meta-data=/dev/nvme1n1           isize=512    agcount=16, agsize=327680 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=1
         =                       reflink=1    bigtime=1 inobtcount=1 nrext64=0
data     =                       bsize=4096   blocks=5242880, imaxpct=25
         =                       sunit=1      swidth=1 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=1 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
ubuntu@ip-172-31-31-47:~$
```

```
realtime =none                   extsz=4096   blocks=0, rtextents=0
ubuntu@ip-172-31-31-47:~$ lsblk -fs
NAME        FSTYPE   FSVER LABEL          UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0       squashfs 4.0                                                           0   100% /snap/amazon-ssm-agent/11797
loop1       squashfs 4.0                                                           0   100% /snap/amazon-ssm-agent/12322
loop2       squashfs 4.0                                                           0   100% /snap/core22/2216
loop3       squashfs 4.0                                                           0   100% /snap/snapd/25202
loop4       squashfs 4.0                                                           0   100% /snap/snapd/25935
loop5       squashfs 4.0                                                           0   100% /snap/core22/2292
nvme1n1     xfs                          e12d541c-c2d3-4c22-be4f-29a50d747028
nvme2n1
nvme0n1p1   ext4     1.0   cloudimg-rootfs 12f6092b-46d6-4b4c-8ac7-d76c4d62dde1    4.2G    37% /
└─nvme0n1
nvme0n1p14
└─nvme0n1
nvme0n1p15 vfat      FAT32 UEFI           BDC4-D6EC                              98.2M     6% /boot/efi
└─nvme0n1
nvme0n1p16 ext4      1.0   BOOT           3ad5394a-0d50-4795-835e-1b2da7ac508d  664.2M    18% /boot
└─nvme0n1
ubuntu@ip-172-31-31-47:~$
```

Step 8: Create a Mount Directory

mkdir /mnt/siddu

```
ubuntu@ip-172-31-31-47:~$ sudo mkdir /mnt/siddu
ubuntu@ip-172-31-31-47:~$
```

Step 9: Mount the Volume

mount /dev/xvdf1 /mnt/archana

Verify:

df -h

```
ubuntu@ip-172-31-31-47:~$ sudo mkdir /mnt/siddu
ubuntu@ip-172-31-31-47:~$ sudo mount /dev/nvme1n1 /mnt/siddu
ubuntu@ip-172-31-31-47:~$ df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root        6.8G  2.5G  4.3G  37% /
tmpfs            458M     0  458M   0% /dev/shm
tmpfs            183M  896K  182M   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
efivarfs         128K  4.4K  119K   4% /sys/firmware/efi/efivars
/dev/nvme0n1p16  881M  155M  665M  19% /boot
/dev/nvme0n1p15  105M  6.2M   99M   6% /boot/efi
tmpfs             92M   12K   92M   1% /run/user/1000
/dev/nvme1n1      20G  425M   20G   3% /mnt/siddu
ubuntu@ip-172-31-31-47:~$
```

Step 10: Make the Mount Persistent

Get the UUID of the volume:

blkid /dev/xvdf1

```
ubuntu@ip-172-31-31-47:~$ sudo blkid /dev/nvme1n1
/dev/nvme1n1: UUID="e12d541c-c2d3-4c22-be4f-29a50d747028" BLOCK_SIZE="512" TYPE="xfs"
ubuntu@ip-172-31-31-47:~$
```

i-05be5029351191b8c

Edit fstab:

nano /etc/fstab

```
  GNU nano 7.2                                                    /etc/fstab
LABEL=cloudimg-rootfs   /        ext4   discard,commit=30,errors=remount-ro     0 1
LABEL=BOOT      /boot   ext4    defaults        0 2
LABEL=UEFI      /boot/efi       vfat    umask=0077      0 1

                                                [ Read 3 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket  M-Q Previous
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^Q Where Was    M-W Next
```

i-05be5029351191b8c

Add this line at the end:

UUID=<UUID-value> /mnt/archana xfs defaults,nofail 0 0

Save and exit.

```
  GNU nano 7.2                                                    /etc/fstab *
LABEL=cloudimg-rootfs   /        ext4   discard,commit=30,errors=remount-ro     0 1
LABEL=BOOT      /boot   ext4    defaults        0 2
LABEL=UEFI      /boot/efi       vfat    umask=0077      0 1
UUID=e12d541c-c2d3-42cc-be4f-29a50d747028 /mnt/siddu xfs defaults,nofail 0 2
^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute     ^C Location    M-U Undo       M-A Set Mark   M-] To Bracket  M-Q
```

Test fstab:

mount -a

Step 11: Verify Persistence

1. Reboot the instance:

reboot

2. Reconnect and check:

df -h

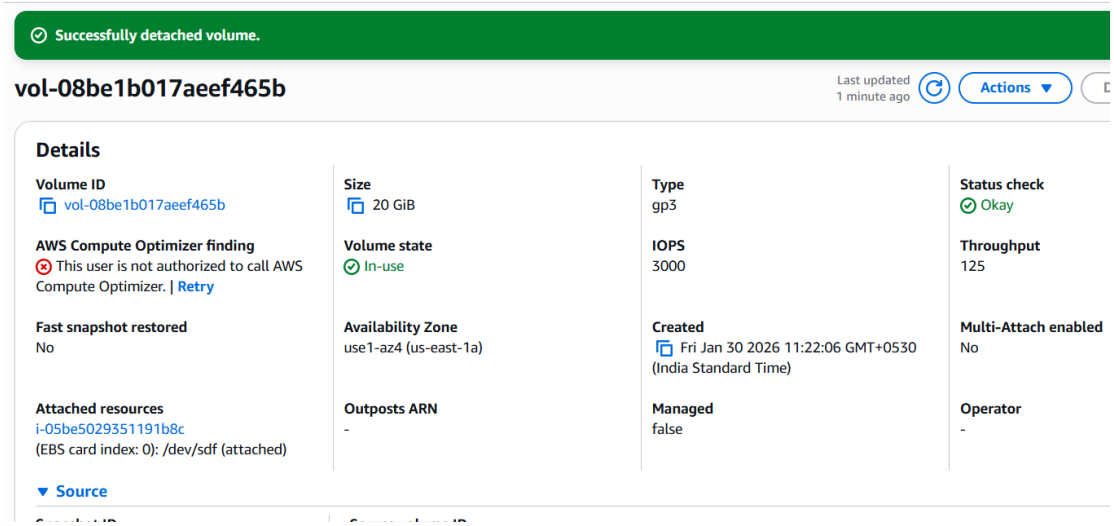
→ The volume should mount automatically.

```
ubuntu@ip-172-31-31-47:~$ sudo mount -a
ubuntu@ip-172-31-31-47:~$ df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root        6.8G  2.6G  4.2G  39% /
tmpfs            458M     0  458M   0% /dev/shm
tmpfs            183M  928K  182M   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
efivarfs         128K  4.4K  119K   4% /sys/firmware/efi/efivars
/dev/nvme0n1p16  881M  155M  665M  19% /boot
/dev/nvme0n1p15  105M  6.2M   99M   6% /boot/efi
tmpfs             92M   12K   92M   1% /run/user/1000
/dev/nvme1n1      20G  425M   20G   3% /mnt/siddu
ubuntu@ip-172-31-31-47:~$
```

Step 12: Detach and Reattach Volume (Data Recovery Scenario)

Detach the Volume

- **EC2 → Volumes → Select volume**
- **Actions → Detach volume**

- 

Attach to Another EC2 Instance (Same AZ)

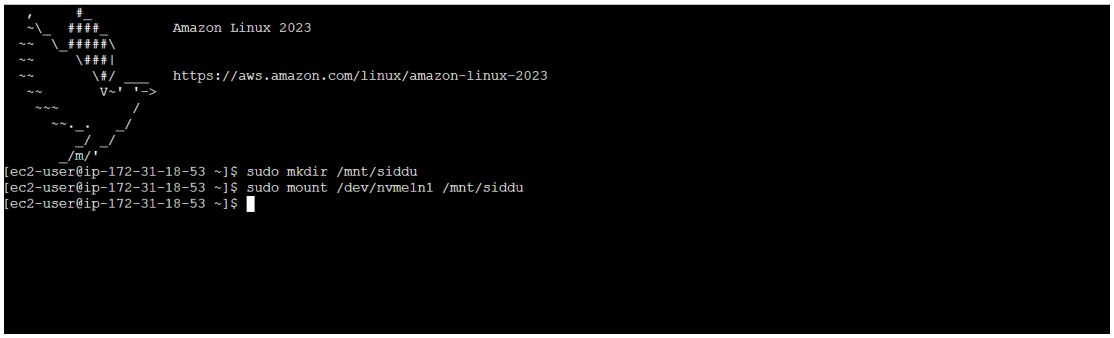
1. Attach the volume to a new instance
2. Log in to the new instance
3. Create mount directory:

mkdir /mnt/archana

4. Mount the existing partition:

mount /dev/nvme1n1p1 /mnt/archana

✅ All previously stored data will be available.



Summary Flow

Create Volume → Attach → lsblk → fdisk → mkfs → mkdir → mount → fstab → reboot

---

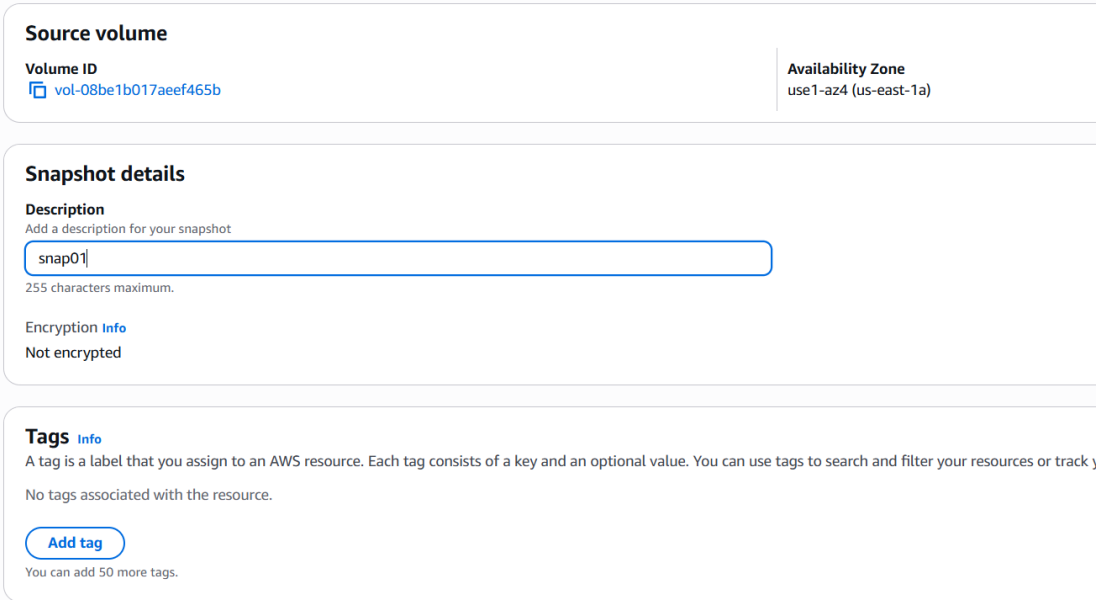
3. Create a snapshot of the attached EBS volume and use it to create and attach a new volume to an EC2 instance in another AWS region.

Attaching Volumes across regions of EC2 using the snapshot

To create a snapshot of an EBS volume and attach it to an instance in another region in AWS, you'll need to follow these steps:

Step 1. Create a Snapshot:

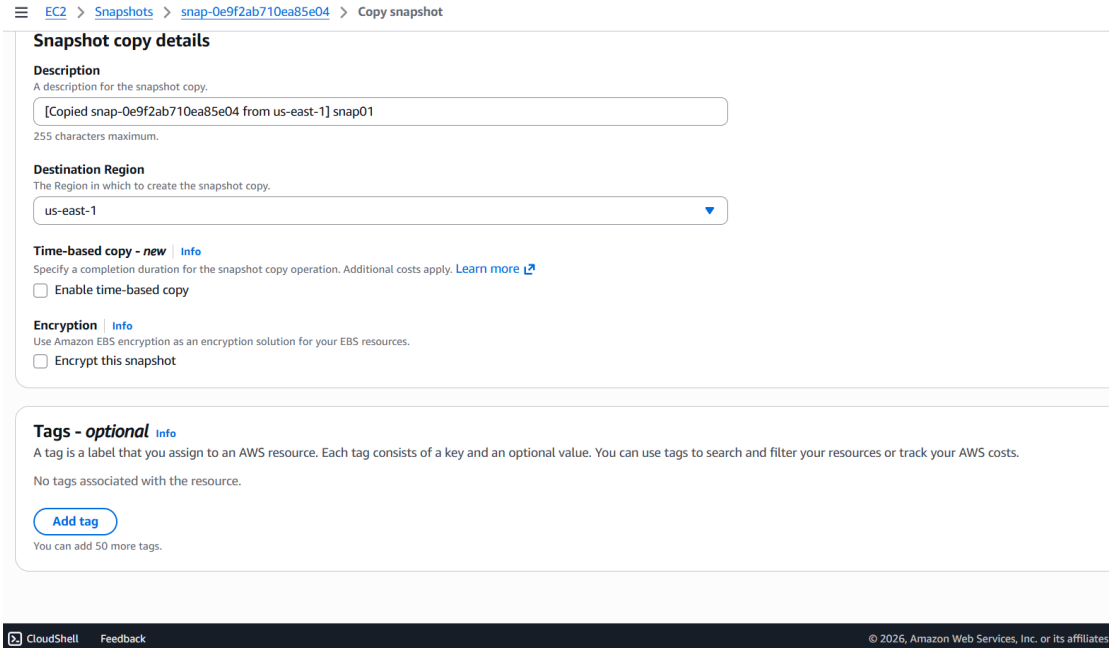
a. Sign in to the AWS Management Console.
b. Navigate to the EC2 Dashboard.
c. Click on "Volumes" in the left-hand navigation pane.
d. Select the EBS volume you want to create a snapshot of.
e. Click on the "Actions" dropdown menu above the volume list and select "Create Snapshot."
f. Provide a name and description for the snapshot.
g. Click on the "Create Snapshot" button to initiate the snapshot creation process.



Step 2.Copy the Snapshot to Another Region:

a. Once the snapshot is created, go to the "Snapshots" section in the EC2 Dashboard.
b. Select the snapshot you just created.
c. Click on the "Actions" dropdown menu above the snapshot list and select "Copy Snapshot."
d. Choose the destination region where you want to copy the snapshot.

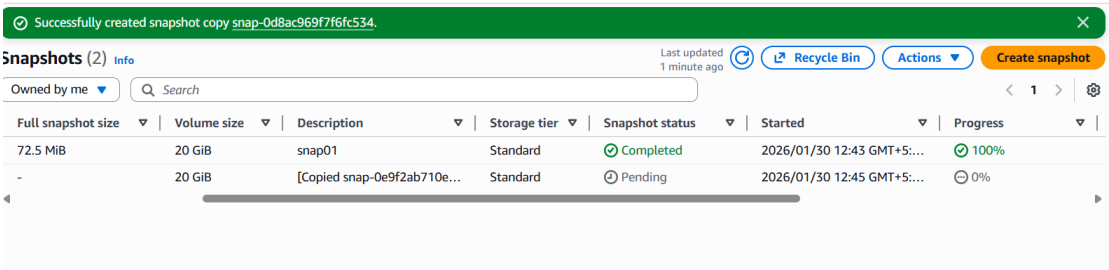
e. Click on the "Copy Snapshot" button to initiate the copy process. This may take some time depending on the size of the snapshot and the network speed.



Step 3. Monitor the Snapshot Copy Progress:

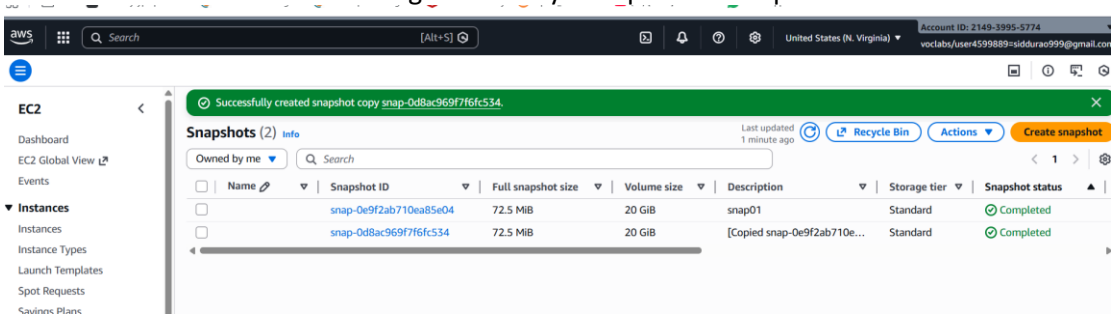
a. You can monitor the progress of the snapshot copy by navigating to the "Snapshots" section in the EC2 Dashboard of the source region.

b. Look for the snapshot you copied and check its status. It will change from "pending" to "completed" once the copy process is finished.



Step 4.Switch to the Destination Region:

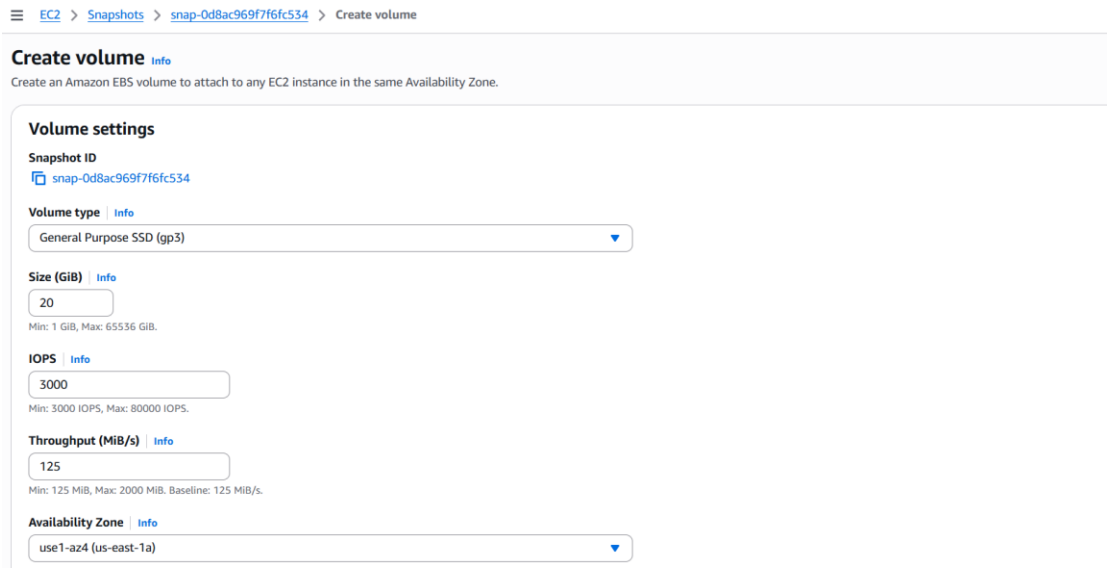
a. Use the region selector in the top-right corner of the AWS Management Console to switch to the destination region where you copied the snapshot.



Step 5. Create a Volume from the Snapshot:

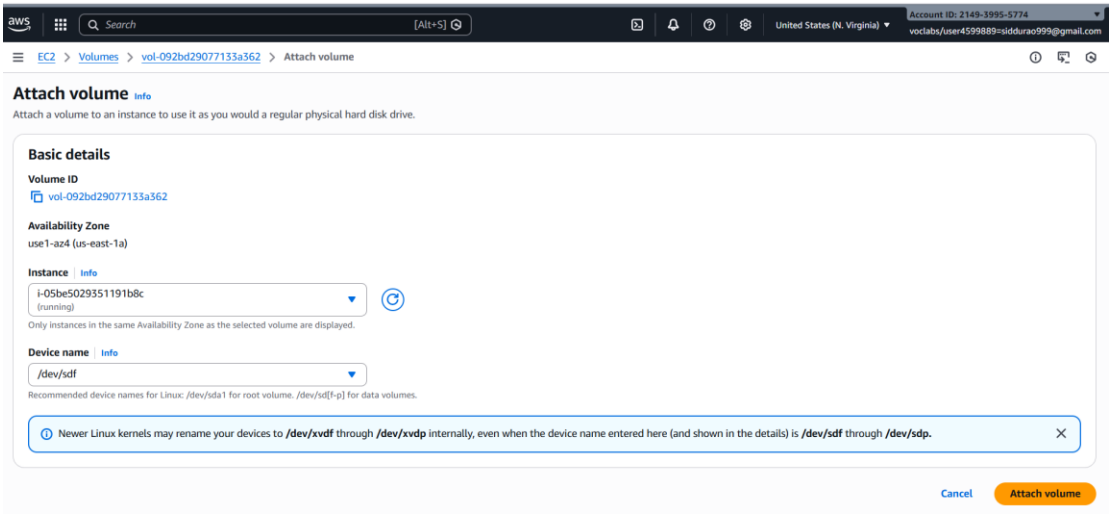
a. In the EC2 Dashboard of the destination region, go to the "Snapshots" section.

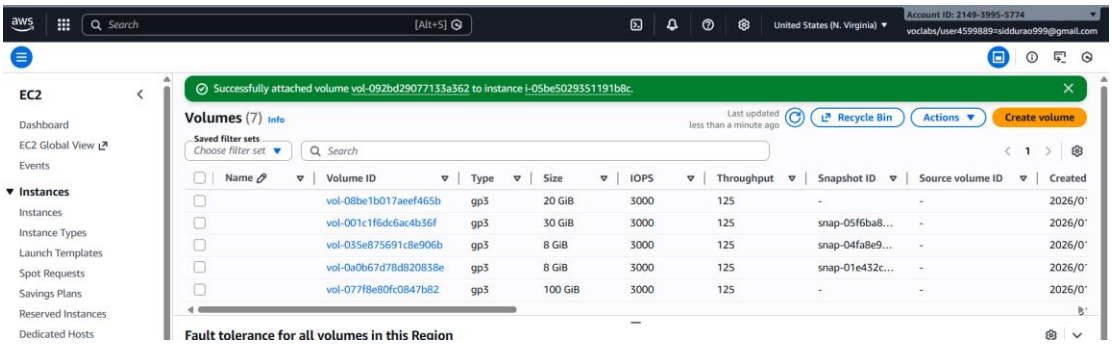
b. Find the snapshot you copied from the source region.
c. Click on the snapshot, then click on the "Actions" dropdown menu and select "Create Volume."
d. Configure the volume settings, such as volume type, size, and availability zone.
e. Click on the "Create Volume" button to create the volume from the snapshot.



Step 6. Create the ec2 instance &Attach the Volume to an Instance:
a. Once the volume is created, navigate to the "Volumes" section in the EC2 Dashboard.
b. Find the newly created volume and select it.
c. Click on the "Actions" dropdown menu and select "Attach Volume."
D the EC2 instance to which you want to attach the volume and specify the device name.
e. Click on the "Attach" button to attach the volume to the instance.

We have created a snapshot of an EBS volume, copied it to another region, created a volume from the snapshot, and attached it to an instance in the destination region.





VIVA :

1.Your database server crashed. Data must be recovered fast. What do you do?
Restore the database from the latest EBS snapshot or automated backup.
Attach the restored volume to a new or existing EC2 instance and restart the database service.

2.EC2 is deleted accidentally—how do you recover data?
Launch a new EC2 instance in the same Availability Zone.
Attach existing EBS volumes or restore data from EBS snapshots.

3.Root volume deleted with EC2. How do you prevent this next time?
Disable the *Delete on Termination* option for the root EBS volume.
This ensures the root volume is preserved even if EC2 is terminated.

4.Disk is full in production EC2. What will you do?
Increase the EBS volume size and extend the file system.
Alternatively, attach a new EBS volume and move data.

5.You want to migrate EC2 to another AWS region. How do you do it?
Create an AMI or EBS snapshot of the instance.
Copy it to the target region and launch a new EC2 there.

6.What happens to EBS when EC2 is stopped?
EBS volumes remain attached and data is fully preserved.
Only compute resources are stopped, not storage.

7.What happens to EBS when EC2 is terminated?
The root EBS volume is deleted by default.
Additional EBS volumes remain unless configured to delete.

8.Application requires 20,000 IOPS. Which EBS volume will you choose?
Provisioned IOPS SSD (io1 or io2) is used for high-performance workloads.
It supports very high and consistent IOPS.

9.Can one EBS volume attach to multiple EC2 instances?
Yes, using EBS Multi-Attach.
It works only with io1/io2 volumes in the same Availability Zone.

10.**You deleted an EBS volume by mistake. How can you recover it?**
Restore the volume from the most recent EBS snapshot.
Without a snapshot, the data cannot be recovered.

11.What happens if a snapshot used by a volume is deleted?
The running volume is not affected and continues to work.
Only future restores from that snapshot are not possible.

12.Where are EBS snapshots stored?
EBS snapshots are stored in Amazon S3.
They are managed internally by AWS and highly durable.